Journal of Anxiety Disorders 25 (2011) 972–977

Contents lists available at ScienceDirect

Journal of Anxiety Disorders

Attention training in socially anxious children: A multiple baseline design analysis

Maria J.W. Cowart*, Thomas H. Ollendick[1]

Virginia Polytechnic Institute and State University, Child Study Center, Department of Psychology, Blacksburg, VA 24060, United States

ARTICLE INFO

Article history:
Received 14 January 2011
Received in revised form 8 June 2011
Accepted 8 June 2011

Keywords:
Social anxiety disorder
Children
Attention
Treatment
Multiple-baseline design

ABSTRACT

Current evidence has established an association between anxiety and attentional threat biases. Emerging research suggests these attentional biases may play a causal role in anxiety development. Clinical researchers have begun to develop treatments specifically designed to address these attentional processes. As such, Attention training (ATT) has been shown to be effective in the treatment of anxiety in adults. The current study represents an early attempt to implement ATT to treat social anxiety disorder (SOC) in children. Two boys meeting criteria for DSM-IV SOC participated in the study, along with their parents. Both boys received 10, 10-min sessions of ATT. A multiple-baseline design was used. Following treatment, both boys evidenced reductions in social anxiety. The current study provides preliminary evidence regarding the utility of ATT for treatment of childhood SOC. Future research is needed to further examine the use of this treatment strategy with youth and to explore the mechanisms of change.

Anxiety disorders affect 10–20% of children and adolescents, cause significant distress and impairment, and may have a chronic course (see Costello, Egger, & Angold, 2004 for review). Currently, cognitive behavioral therapy (CBT) is considered the treatment of choice for such disorders, in both children and adults. CBT has been shown to be effective for 50–70% of anxious children and adults (Barlow, 2001; Otto, Behar, Jasper, & Hofman, 2009) and 60–70% of anxious children (Kendall et al., 2005; Seligman & Ollendick, 2011; Walkup et al., 2008). These moderate remission rates leave room for refinement of our current treatments and/or development of new treatment strategies to address anxiety.

One area that has shown promise with regard to new treatments is attentional processing. A large body of physiological and psychological evidence has established an association between anxiety and biases of attention (see Bar-Haim, Lamy, Pergamin, Bakermans-Kranenburg, & van Ijzendoorn, 2007, for a review). Specifically, anxious individuals demonstrate a threat perception bias, such that they are vigilant to threatening stimuli and notice such stimuli earlier than non-anxious individuals. Further, those with an anxiety disorder may have more difficulty than those without an anxiety disorder disengaging their attention from threat cues (Fox, Russo, Bowles, & Dutton, 2001).

Indeed, attentional biases to threat have emerged as playing a prominent role in the etiology and maintenance of anxiety disorders (Eldar, Ricon, & Bar-Haim, 2008; Eysenck, 1997; Lonigan, Vasey, Phillips, & Hazen, 2004; Williams, Watts, MacLeod, & Matthews, 1997). For example, Eldar and colleagues (2008) examined these biases in a group of normal children. One group of children was trained to attend to threat cues while the other group was trained to attend away from threat cues. Both groups of children were then exposed to a stress induction; however, only children who were trained to attend to threat exhibited significant elevations in anxiety. If attentional biases to threat serve a causal function in the development of anxiety disorders, or at least serve to maintain them, treatment could address these biases. Currently, as noted, treatment of choice for anxiety disorders is CBT. One of the primary goals of CBT is to reduce or eliminate anxiety-related cognitive biases. Often, this is accomplished using cognitive restructuring, involving the identification of and recording of maladaptive cognitions, as well as the challenging of those cognitions with competing evidence and more accurate cognitions (cf. Barlow, 2001; Prins & Ollendick, 2003). Although these techniques aid in the treatment of anxiety, some have argued they are not sufficient to address the attentional biases that seem to be an integral part of the development and maintenance of these disorders (Papageorgiou & Wells, 1998).

Recently, a novel approach, attention training (ATT), has sought to address anxiety disorders directly through the remediation of attentional biases (Bar-Haim, 2010; Mohlman, 2004). While

* Corresponding author. Tel.: +1 540 231 8276; fax: +1 540 231 8193.
E-mail addresses: mjw@vt.edu (M.J.W. Cowart), tho@vt.edu (T.H. Ollendick).
[1] Tel.: +1 540 231 6451; fax: +1 540 231 8193.

0887-6185/$ – see front matter
doi:10.1016/j.janxdis.2011.06.005

methodological differences exist across studies (e.g., number of training sessions, length of sessions), some basic strategies are common across all ATT protocols (Mohlman, 2004). Typically, sessions are brief (e.g., 10–30-min) and involve experimental tasks designed with neutral stimuli as targets and anxiety-related stimuli as distracters. The goal of such training is to learn the tendency to disengage attention from anxiety-related stimuli in the environment, in favor of attention to neutral stimuli. ATT has been successfully utilized with a variety of anxious adult populations. Although ATT has most commonly been employed with socially anxious adults (e.g., Amir, Beard, Burns, & Bomyea, 2009; Klumpp & Amir, 2010; MacLeod & Bridle, 2006; Schmidt, Richey, Buckner, & Timpano, 2009), it has also been used to treat individuals with generalized anxiety disorder (Amir, Beard, Burns, et al., 2009), pathological worry (Hazen, Vasey, & Schmidt, 2009), subclinical obsessive-compulsive disorder (OCD; Najmi & Amir, 2010), high trait levels of anxiety (Rutherford, MacLeod, & Campbell, 2004), and those undergoing stressful life events (MacLeod & Bridle, 2006).

In one of the first randomized controlled trials of attention training, Amir and colleagues (2009) compared the effects of attention training versus an attention control task on social anxiety in a sample of 48 adults meeting criteria for a principal diagnosis of social anxiety disorder. Following eight, twice-weekly 20-min sessions of ATT, participants displayed improvements in attentional biases as well as social anxiety, as compared to those in the control condition. Attentional biases were assessed at pre- and post-treatment with a Posner task, such that increased response latency indicated difficulty disengaging attention from threatening stimuli. At post-treatment assessment, those who received attention training displayed somewhat faster response times on the task, while the response times of the control group did not change over time. With regard to social anxiety, participants receiving attention training endorsed fewer symptoms of social anxiety and less functional impairment than the control group on a variety of self-report measures. Moreover, these gains were maintained at 4-month follow-up. That same year, Schmidt and colleagues (2009) reported similar results to Amir et al. in a sample of 36 socially anxious adults. Following treatment with eight, twice-weekly sessions of ATT, 72% of participants no longer met criteria for a diagnosis of social anxiety disorder, compared to 11% of those in the control group. As in Amir et al., gains were maintained at 4-month follow-up.

Despite success of ATT in addressing anxiety in adults, this approach has not yet been applied to anxious children or adolescents. The current report represents a preliminary attempt to treat social anxiety disorder with attention training in two children using a multiple baseline design. It was expected that ATT would reduce both social anxiety and related attentional biases toward threat.

1. Method

1.1. Participants

Prospective participants were recruited from the community through use of flyers, radio advertisement, and referrals from a university-based mental health clinic. These prospective participants were initially screened using a telephone interview. The telephone interview was administered to parents, and briefly assessed for symptoms of social anxiety disorder, as well as any exclusionary criteria. Exclusionary criteria included (a) meeting full diagnostic criteria for an autism spectrum disorder, (b) current and acute psychotic symptoms, (c) estimated IQ less than 80, or (d) current suicidal or homicidal ideation. Of 7 participants who completed the telephone screening, 3 met inclusion criteria and were entered into the treatment protocol. However, one participant dropped out mid-treatment due to family relocation.

The two children who participated, Adam and Seth (pseudonyms), were 8 and 9 years old, respectively, and both were Caucasian. Adam was in 3rd grade, while Seth was in 4th grade. Both boys attended regular elementary schools, and Seth was receiving services under the gifted designation. Neither child had received previous mental health services.

1.2. Measures

1.2.1. Anxiety Disorders Interview Schedule for Children, Parent Version (ADIS-P; Silverman & Albano, 1996)

The ADIS-P is a semistructured diagnostic interview designed to assess childhood anxiety disorders, as well as other related disorders (e.g., ADHD), as reported by the parents. For purposes of the current study, the ADIS-P was administered by trained graduate students in an APA-approved clinical psychology program. Following the interview, clinicians assign severity ratings for each diagnosis, on a scale from 0 to 8. Utilizing this system, children received a clinical diagnosis if the parent clinician assigned the diagnosis with a severity rating of 4 or above (see Silverman & Albano, 1996). The ADIS-P has demonstrated adequate test-retest parent diagnoses (kappas of .65–1.00; Silverman, Saavedra, & Pina, 2001) and inter-rater agreement in our Center (Grills & Ollendick, 2003).

1.2.2. Spence-Children's Anxiety Scale, Parent Version (SCAS; Spence, 1998)

The SCAS is a parent-report questionnaire assessing anxiety symptoms in children. Parents are asked to respond to each of 38 items by indicating the frequency with which their child experiences each symptom, with ratings ranging from Never (scored 0) to Always (scored 4). The 38 items contribute to six subscales, as follows: social phobia, separation anxiety, panic attack/agoraphobia, obsessive-compulsive disorder, generalized anxiety disorder, and physical injury fears. The SCAS has demonstrated good internal consistency with an overall coefficient alpha of .89 (.74 for the social phobia subscale). The entire SCAS was administered at baseline, end of baseline, and post-treatment. The SCAS Social Phobia subscale was administered at each weekly session to monitor treatment progress.

1.2.3. Attention assessment

A computerized pictorial dot-probe paradigm was utilized to identify attentional biases to threat. The task was designed by Amir and colleagues, and is similar to those utilized in previous investigations (e.g., Amir, Beard, Taylor, et al., 2009). Task stimuli were comprised of standardized facial images developed by Matsumoto and Ekman (1988), displaying threatening (disgust, anger) and neutral expressions. Images were paired, such that the same person was pictured in each image. Image pairs were presented with one image at the top half of the computer screen, while the other image appeared on the bottom half of the screen.

During the task, participants first saw a small fixation cross at the center of the computer screen for approximately 500 ms. Following the presentation of the fixation cross, a pair of facial images was presented for 500 ms. After facial stimuli disappeared, a probe (either the letter E or the letter F) appeared in the location of one of the two faces. Participants were instructed to decide if the letter was an E or an F by pressing the corresponding button (left or right) on the computer mouse. The probe remained on the screen until participants responded, after which the next trial began. Participants were told that it was important to complete the task as quickly as possible without sacrificing accuracy. Participants completed a total of 256 trials, including various combinations of probe type (E or F), facial stimuli type (anger, disgust or neutral) and facial stimuli gender (women or men). The probe task is designed to reflect attentional processes such that participants should be faster responding to probes which occur in the area of the previously presented stimulus to which they were attending. Thus, faster mean response times to threat stimuli versus neutral stimuli indicate an attentional bias to threat.

2. Procedure

Participants and their parents provided written informed consent and assent. A pre-treatment session lasted approximately 2 h, and was conducted by graduate students enrolled in an APA-approved program in clinical psychology. One clinician completed assessment with the mother, while a second clinician completed assessment with the child. Following the pre-treatment assessment, participants completed a randomly assigned number of baseline assessments to fulfill criteria for a multiple baseline design study. A multiple baseline design was chosen because it allows for a level of experimental control to demonstrate change as a result of treatment. Adam and his mother were randomly assigned to three baseline sessions, whereas Seth and his mother were randomly assigned to complete five baseline sessions. Baseline assessments lasted approximately 30 min, and consisted of completion of the attention assessment (dot-probe task) and the weekly SCAS Social Phobia questionnaire.

Following baseline assessment, both participants were enrolled in treatment. Both children were treated by the first author, who was a masters-level graduate student in an APA-approved program in clinical psychology, and supervised by the second author. Treatment involved 10, twice-weekly sessions which were approximately one half-hour in length. Approximately 10 min of each session were devoted to the administration of the attention training task. This task was similar to the dot-probe task described above, with several modifications (see Amir, 2009 for a more detailed description of the task). First, the treatment task consisted of 160 trials, in which a probe replaced a neutral stimulus in 128 of the trials (80% of trials); in the remaining 32 “filler” trials, the probe replaced the threatening stimulus (20% of trials). For purposes of the treatment task, 8 angry facial stimuli and 8 disgust facial stimuli were utilized and paired with 16 neutral facial stimuli. As in the dot-probe task, the pairs consisted of two pictures of the same person. The remainder of the half-hour session was comprised of basic rapport- and relationship-building with the therapist, as well as completion of questionnaires.

Approximately 2 weeks after treatment, the children and their mothers underwent post-treatment assessment. This session was similar to the pre-treatment assessment, and was again administered by trained graduate students in clinical psychology.

3. Results

3.1. Social anxiety

3.1.1. Diagnostic interviews

Based on the ADIS-P at pre-treatment, Adam was assigned a diagnosis of social anxiety disorder with a clinician severity rating of 6 (on a 0–8-scale). Following treatment, the ADIS-P was re-administered to Adam’s mother. At that time, Adam’s social anxiety disorder was assigned a clinician severity rating of 3, indicating subclinical levels of social anxiety. Qualitatively, Adam’s mother described several recent incidents in which he was able to engage in behaviors he was unable to complete at pre-treatment (e.g., order his own food in a restaurant, converse with adult friends of the family).

Similarly, Seth was assigned a diagnosis of social anxiety disorder at pre-treatment based on the ADIS-P (clinician severity rating of 6). At post-treatment, the clinician severity rating of Seth’s social anxiety disorder was a 2, suggesting subclinical social anxiety disorder. Seth’s mother also described him as more socially confident following treatment.

3.1.2. Self-report questionnaires

Adam’s mother completed the SCAS with regard to Adam’s anxiety symptoms. She reported a total score of 40 at pre-treatment, with a score of 15 at post-treatment. Similarly, Seth’s mother reported a score of SCAS score of 34 for Seth at pre-treatment, with a score of 9 at post-treatment. Both sets of scores indicate a reduction in overall anxiety from pre- to post-treatment and to near normative levels of overall anxiety on the Spence scale.

Parent-reported scores on the SCAS-SP for both children across the study are shown in Fig. 1. Across three baseline sessions, Adam’s scores were somewhat variable, ranging from 6 to 10. This inconsistent pattern was noted for the first three sessions of treatment as well. At the fourth treatment session, however, Adam’s SCAS-SP score dropped to 4, where it remained stable throughout the remainder of treatment. At post-treatment, Adam’s mother reported a SCAS-SP score of 2, suggesting very little social anxiety.

Seth’s scores across five baseline sessions also ranged from 6 to 10, and remained stable at 9 for the first three treatment sessions. However, beginning with the fourth treatment session, Seth’s SCAS-SP scores dropped to 5, where they remained relatively stable throughout the remainder of treatment. At post-treatment, Seth’s mother reported a SCAS-SP score of 1, indicating minimal social anxiety.

3.2. Attentional bias

Before calculating attentional bias scores, trials with errors or extreme response times (<100 ms or >3000 ms) were removed. This resulted in the removal of 10% of trials for Adam, and 5% of trials for Seth. Adam’s mean overall response time across trials was 768 ms, and Seth’s mean overall response time was 804 ms. As in previous studies (e.g., Mogg & Bradley, 1999; Pine et al., 2005), attentional bias scores were calculated by subtracting mean response times to threat stimuli from mean response times to neutral stimuli. As noted above, faster response times to a probe indicate that attention was allocated to a stimulus in the same position on a given trial. Thus, positive scores indicate an attentional bias toward threat, while negative scores suggest a bias toward neutral stimuli and away from threat.

Unexpectedly, at pre-treatment, neither Adam nor Seth demonstrated a bias toward threat. Rather, both boys displayed an attentional bias toward neutral stimuli. Adam’s score suggested a negligible neutral bias (−13.83 ms), while Seth’s score indicated a greater preference for neutral stimuli (−75.13 ms). Because neither boy demonstrated an attentional bias toward threat at pre-treatment, there was no opportunity for positive change on this measure. Therefore, post-treatment data on this measure were not examined.

4. Discussion

Two boys with social anxiety disorder displayed significant reductions in social anxiety as a result of treatment with attention training. Across 5 weeks of treatment (10 sessions), the boys displayed somewhat similar patterns of anxiety reduction, as reported

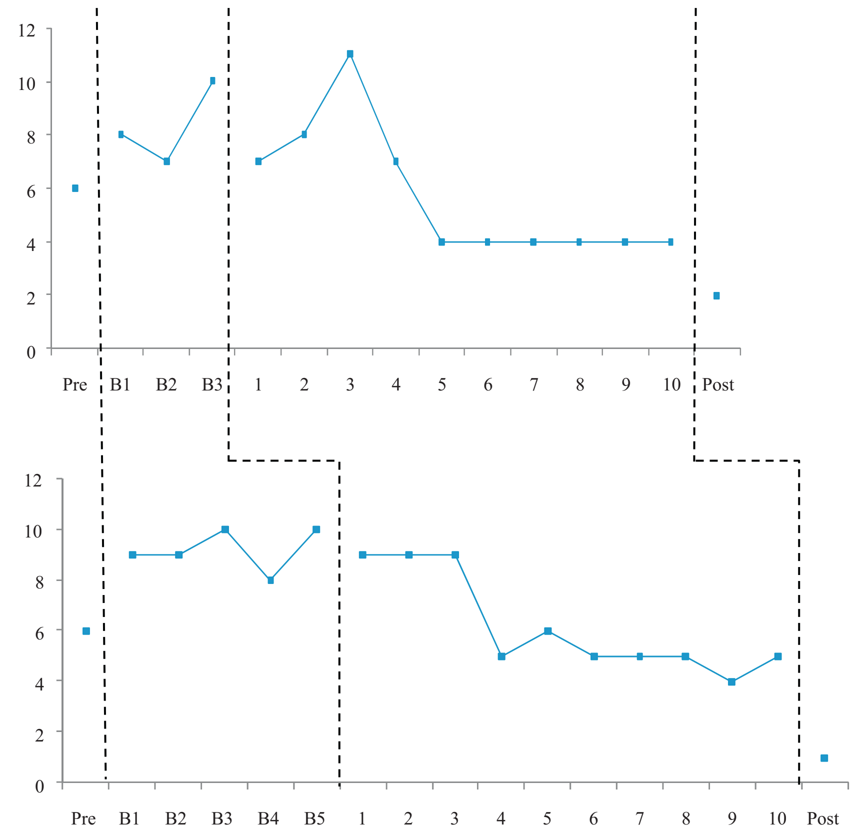

Fig. 1. SCAS-SP scores for both children over the course of treatment.

by their mothers on self-report questionnaires. Specifically, for both boys, reduction in social anxiety was most evident during the fourth session of attention training, with gains maintained after that point. Additional reduction in anxiety was noted for both boys at a post-treatment assessment, approximately 2 weeks following treatment. Consistent with scores on questionnaire measures, the boys displayed moderate to severe social anxiety disorder as reported by their mothers at pre-treatment. However, at post-treatment, neither boy met criteria for a diagnosis of social anxiety disorder. Thus, the current study provides preliminary evidence for utility of attention training in the treatment of childhood social anxiety disorder.

With regard to attentional biases, however, results were counter to expectations. Neither boy displayed the expected attentional bias toward threat at pre-treatment (i.e., faster response times to probes in the area of a threatening stimulus). Rather, the boys displayed very little difference in their response times to threatening stimuli as compared to neutral stimuli. Moreover, the negligible difference in response times to the two different types of stimuli were in the unexpected direction, such that the boys were marginally faster in responding to neutral stimuli as compared to threatening stimuli. Thus, the boys were almost equally likely to attend to a neutral face as to a threatening face. Therefore, reduction in biased attention toward threat is unlikely to be responsible for the treatment gains for these children. This is in contrast to previous research that has implicated attentional biases to threat as a causal and maintaining factor in anxiety at least for adults (e.g., MacLeod, Rutherford, Campbell, Ebsworthy, & Holker, 2002), and has suggested that change in attentional biases mediates treatment outcome in attention training with adults (Amir, Beard, Burns, et al., 2009).

Consistent with our findings, some recent research has begun to call into question the relationship between social anxiety disorder and attentional biases to threat. For example, in a sample of socially anxious adults, Sposari and Rapee (2007) failed to demonstrate an attentional bias toward threatening faces, although socially anxious individuals displayed a preference for attending to facial stimuli rather than object stimuli under conditions of social threat. Other studies have suggested a suppression of attentional threat bias under conditions of threat. For example, Amir and colleagues (1996) demonstrated a suppression of the Stroop interference effect for socially anxious participants under conditions of an impending speech task. Thus, participants in the current study may have found the therapy sessions socially threatening (e.g., because of demands for attending to the facial stimuli and perhaps to the assessor as well), and therefore, their attentional bias may have been suppressed.

Further, some inconsistency in research findings with regard to attentional biases may be due to individual differences, such that some individuals with social anxiety show attentional bias toward threat, while others do not. Indeed, Bar-Haim (2010) reports that the distribution of threat bias in anxious individuals is relatively normal, indicating that large numbers of individuals do not show the bias.

Despite their lack of attentional bias to threat, the children in this study benefitted from attention training with regard to anxiety reduction as evidenced in the multiple baseline design. Neither child demonstrated change until the treatment was implemented. However, at this time, the mechanism of treatment remains unclear. If not attentional bias, it may be that the attention training task produced effects on attentional control (Bar-Haim, 2010). That is, the attention training may have improved the children's ability to efficiently disengage attention from one stimulus to engage with another, making attentional processing more fluid. Indeed, there is evidence that such attentional control serves as a protective factor against development of anxiety (Lonigan & Vasey, 2009).

Alternately, it is possible that that simply interacting with a therapist during sessions produced change in social anxiety for these children. Current evidence suggests that exposure-based therapy is a highly effective treatment for social anxiety disorder (Hofmann & Otto, 2008). In the current study, the brief rapport-building period of each session may have served as an exposure activity for these socially anxious children and was responsible for the observed changes in social anxiety. Unfortunately, the design of the current study does not allow for definitive conclusions about the active ingredients of the treatment, including the rapport-building portion of sessions.

Of course, results of the current study may have also been a result of client expectations and placebo effects (Rosenthal & Frank, 1956). The current study did not include a control group and therefore, both families knew they were receiving active treatment. This may have contributed to positive outcome expectancies and affected parents' reports of their children's symptoms. Moreover, positive treatment expectations may have produced behavioral change in the children, apart from any direct therapeutic effects (Lott & Murray, 1975). However, use of the multiple baseline design in the current study suggests that placebo effects or client expectations are less likely to explain treatment effects since changes were not observed during baseline and only became evident about mid-way through treatment. Further research, including randomized controlled trials with larger samples of children, is needed to determine if the treatment effects of ATT are robust.

Inclusion of only two participants in the current study also limits the conclusions that can be drawn. Children included in the study were similar with regard to a range of demographic features, including age, educational level, ethnicity, and family background. Further, severity of social anxiety disorder did not differ between the two children. Nonetheless, the current study provides preliminary evidence for the effectiveness of ATT with socially anxious children. As noted above, there are no studies yet published regarding the use of ATT with children. This is a major gap in the extant literature. Given the accumulating evidence for the effectiveness of ATT in treating anxious adults, it is critical that this treatment approach be explored further for its potential use with children.

References

Amir, N., Beard, C., Burns, M., & Bomyea, J. (2009). Attention modification program in individuals with generalized anxiety disorder. *Journal of Abnormal Psychology*, *118*(1), 28–33.

Amir, N., Beard, C., Taylor, C. T., Klumpp, H., Elias, J., Burns, M., et al. (2009). Attention training in individuals with generalized social phobia: a randomized controlled trial. *Journal of Consulting and Clinical Psychology*, *77*(5), 961–973.

Amir, N., McNally, R. J., Riemann, B. C., Burns, J., Lorenz, M., & Mullen, J. T. (1996). Suppression of the emotional Stroop effect by increased anxiety in patients with social phobia. *Behaviour Research and Therapy*, *34*, 945–948.

Bar-Haim, Y. (2010). Research review: attention bias modification (ABM): a novel treatment for anxiety disorders. *Journal of Child Psychology and Psychiatry*, *51*(8), 859–870.

Bar-Haim, Y., Lamy, D., Pergamin, L., Bakermans-Kranenburg, M. J., & van Ijzendoorn, M. H. (2007). Threat-related attentional bias in anxious and nonanxious individuals: a meta-analytic study. *Psychological Bulletin*, *133*, 1–24.

Barlow, D. H. (2001). *Clinical handbook of psychological disorders: a step-by-step treatment manual*. New York: Guilford Press.

Costello, E. J., Egger, H. L., & Angold, A. (2004). Developmental epidemiology of anxiety disorders. In: T. H. Ollendick, & J. S. March (Eds.), *Phobic and anxiety disorders in children and adolescents: a clinician's guide to effective psychosocial and pharmacological interventions* (pp. 61–91). New York: Oxford University Press.

Eldar, S., Ricon, T., & Bar-Haim, Y. (2008). Plasticity in attention: implications for stress response in children. *Behavior Research and Therapy*, *46*(4), 450–461.

Eysenck, M. W. (1997). *Anxiety and cognition: a unified theory*. Hove, England: Psychology Press.

Fox, E., Russo, R., Bowles, R., & Dutton, K. (2001). Do threatening stimuli draw or hold attention in subclinical anxiety? *Journal of Experimental Psychology: General*, *130*(4), 681–700.

Grills, A. E., & Ollendick, T. H. (2003). Multiple informant agreement and the Anxiety Disorders Interview Schedule of Parents and Children. *Journal of the American Academy of Child and Adolescent Psychiatry*, *42*, 30–40.

Hazen, R. A., Vasey, M. W., & Schmidt, N. B. (2009). Attention retraining: a randomized clinical trial for pathological worry. *Journal of Psychiatric Research*, *43*(6), 627–633.

Hofmann, S. G., & Otto, M. W. (2008). Cognitive-behavior therapy for social anxiety disorder: evidence-based and disorder-specific treatment techniques. In *Practical clinical guidebooks series*. New York, NY: Routledge/Taylor & Francis Group.

Kendall, P. C., Hudson, J. L., Choudhury, M., Webb, A., & Pimentel, S. (2005). Cognitive-behavioral treatment for childhood anxiety disorders. In: E. D. Hibbs, & P. S. Jensen (Eds.), *Psychosocial treatments for child and adolescent disorders: empirically based strategies for clinical practice* (2nd ed., pp. 47–73).

Klumpp, H., & Amir, N. (2010). Preliminary study of attention training to threat and neutral faces on anxious reactivity to a social stressor in social anxiety. *Cognitive Therapy Research*, *34*, 263–271.

Lonigan, C. J., & Vasey, M. W. (2009). Negative affectivity, effortful control, and attention to threat-relevant stimuli. *Journal of Abnormal Child Psychology*, *37*, 387–399.

Lonigan, C. J., Vasey, M. W., Phillips, B. M., & Hazen, R. A. (2004). Temperament, anxiety, and the processing of threat-relevant stimuli. *Journal of Clinical Child and, Adolescent Psychology*, *33*, 8–20.

Lott, D. R., & Murray, E. J. (1975). The effect of expectancy manipulation on outcome in systematic desensitization. *Psychotherapy: Theory, Research, and Practice*, *12*(1), 28–32.

MacLeod, C., & Bridle, R. (2006). The modification of anxiety through the manipulation of attention bias: evaluation of a 16 day on-line attentional training program. In *Paper presented at the convention of the European Association of Behavioral and Cognitive Therapies* September 2006.

MacLeod, C., Rutherford, E., Campbell, L., Ebsworthy, G., & Holker, L. (2002). Selective attention and emotional vulnerability: assessing the causal basis of their association through the experimental manipulation of attentional bias. *Journal of Abnormal Psychology*, *111*(1), 107–123.

Matsumoto, D., & Ekman, P. (1988). *The Japanese and Caucasian Facial Expression of Emotion (JACFEE) and Neutrals (JACNeuF) [Slides]*. 1600 Holloway Avenue, San Fransisco, CA 94132: Intercultural and Emotion Experience Laboratory, Department of Psychology, San Fransisco State University.

Mogg, K., & Bradley, B. P. (1999). Some methodological issues in assessing attentional biases for threatening faces in anxiety: a replication study using a modified version of the probe detection task. *Behaviour Research and Therapy*, *37*, 595–604.

Mohlman, J. (2004). Attention training as an intervention for anxiety: review and rationale. *The Behavior Therapist*, *27*(2), 37–41.

Najmi, S., & Amir, N. (2010). The effect of attention training on a behavioral test of contamination fears in individuals with subclinical obsessive-compulsive symptoms. *Journal of Abnormal Psychology*, *119*(1.), 136–142.

Otto, M. W., Behar, E., Jasper, J. A. J., & Hofman, S. G. (2009). Combining pharmacological and cognitive behavioral therapy in the treatment of anxiety disorders. In: M. M. Antony, & M. B. Stein (Eds.), *Oxford handbook of anxiety and related disorders* (pp. 429–440). Oxford, England: Oxford University Press.

Papageorgiou, C., & Wells, A. (1998). Effects of attention training on hypochondriasis: a brief case series. *Psychological Research*, *28*(1), 193–200.

Pine, D., Mogg, K., Bradley, B. P., Montgomery, L., Monk, C. S., McClure, E., et al. (2005). Attention bias to threat in maltreated children: implications for vulnerability to stress-related psychopathology. *The American Journal of Psychiatry*, *162*(2), 291–296.

Prins, P., & Ollendick, T. H. (2003). Cognitive change and enhanced coping: missing meditational links in cognitive behavior therapy with anxiety-disordered children. *Clinical Child and Family Psychology Review*, *6*(2), 87–105.

Rosenthal, D., & Frank, J. D. (1956). Psychotherapy and the placebo effect. *Psychological Bulletin*, *53*(4), 294–302.

Rutherford, E. M., MacLeod, C., & Campbell, L. W. (2004). Negative selectivity effects and emotional selectivity effects in anxiety: differential attentional correlates of state and trait variables. *Cognition and Emotion*, *18*(5), 711–720.

Schmidt, N. B., Richey, J. A., Buckner, J. D., & Timpano, K. R. (2009). Attention training for generalized social anxiety disorder. *Journal of Abnormal Psychology*, *118*(1), 5–14.

Seligman, L. D., & Ollendick, T. H. (2011). Cognitive Behavior Therapy for anxiety disorders in children and adolescents. *Psychiatric Clinics of North America*, *20*, 217–238.

Silverman, W. K., & Albano, A. M. (1996). *Anxiety Disorders Interview Schedule for Children*. San Antonio, TX: Psychological Corporation.

Silverman, W. K., Saavedra, L. M., & Pina, A. A. (2001). Test–retest reliability of anxiety symptoms and diagnoses with anxiety disorders interview schedule for DSM-IV: Child and Parent Versions. *Journal of the American Academy of Child and Adolescent Psychiatry*, *40*(8), 937–944.

Spence, S. H. (1998). A measure of anxiety symptoms among children. *Behaviour Research and Therapy*, *36*(5), 545–566.

Sposari, J. A., & Rapee, R. M. (2007). Attentional bias toward facial stimuli under conditions of social threat in socially phobic and nonclinical participants. *Cognitive Therapy and Research*, *31*(1), 23–37.

Walkup, J. T., Albano, A. M., Piacentini, J., Birmhaer, B., Compton, S. N., Sherrill, J. T., et al. (2008). Cognitive behavioral therapy, sertraline, or a combination in childhood anxiety. *The New England Journal of Medicine*, *359*(26), 2753–2766.

Williams, J. M. G., Watts, F. N., MacLeod, C., & Matthews, A. (1997). *Cognitive psychology and emotional disorders* (2nd ed.). Chichester, England: Wiley.